

PROJECT SUBMISSION REPORT

DocExtractor — Universal Layout-Agnostic Question & Diagram Extraction Engine

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Live Web Application:	https://docextractor-iv9b.onrender.com/
Video Demonstration:	https://youtu.be//AebZnGRI44
GitHub Repository:	https://github.com/labhamsharma1633/DocExtractor
Interactive API Docs:	https://docextractor-iv9b.onrender.com/docs

1. Executive Summary

DocExtractor is an automated, layout-agnostic document processing and question extraction platform. It converts complex digital PDFs, scanned documents, and image-based exam papers into structured, high-accuracy question datasets. Traditional PDF text extractors fail on competitive exam papers (such as NEET, JEE, GATE, and UPSC) due to multi-column text interleaving, missing vector/raster diagrams, and false positives from front-page test instructions and candidate forms. DocExtractor solves these core challenges through a **10-Stage Spatial Pipeline**, an intelligent **Negative Region Classifier**, a **Spatial Layout Visual Extractor**, and **Google Gemini 1.5 Flash AI fallback**.

2. Core Problems Addressed & Solutions

Challenge in Real-World PDFs	Legacy Extractor Flaw	DocExtractor Universal Solution
Two-Column Layouts	Reads horizontally across columns, interleaving sentences.	Spatial Column Slicing: Splits blocks by midline geometry (mid_x) and orders columns sequentially.
Exam Instructions & Forms	Captures OMR rules and candidate forms as questions.	Negative Region Classifier: Filters instructions, candidate metadata, and commands with 0% false positives.
Scientific Diagrams	Diagrams discarded or replaced with page watermarks.	Spatial Visual Extractor: Filters 320x320 watermarks; crops vector/raster diagrams inside question bbox.
Single-Line Options	(1) 10 N (2) 20 N (3) 30 N (4) 0 merged into one line.	Horizontal Option Parser: Normalizes and extracts side-by-side options into distinct keys and values.
Scanned & Noisy Papers	Character degradation causes regex parsers to break.	Hybrid Architecture: Tesseract OCR + Google Gemini 1.5 Flash with strict Pydantic JSON schemas.

3. Verification & Live Document Audit Results

The complete universal extraction pipeline was evaluated on a 24-page competitive exam paper (**Practice Test 02 Test Paper Yakeen 2.0 NEET**):

Metric	Evaluation Result
Total Pages Processed	24 Pages
Total Questions Extracted	196 Questions (Covering Q1 through Q180 cleanly)
Overall Extraction Confidence	87% (Status: COMPLETED)

Instruction False-Positive Rate	0.0% (100% suppression of rules/metadata)
Diagrams Captured & Attached	Q14, Q17, Q21, Q30, Q43, Q172 (Vector crops & raster diagrams)
Automated Test Suite	29 Passed (100% pass rate in pytest)

4. Technology Stack

- **Backend Framework:** Python 3.11, FastAPI, Uvicorn, Pydantic v2, SQLAlchemy ORM
- **Document Processing & OCR:** PyMuPDF (Fitz), Pillow, PyPDF, Tesseract OCR
- **Artificial Intelligence (AI):** Google Gemini 1.5 Flash (Structured JSON Pydantic Schema)
- **Database & Persistence:** SQLite (Dev) / PostgreSQL (Production), Local file storage
- **Frontend Architecture:** HTML5, Modern Vanilla CSS3 (Dark Mode / Glassmorphism), ES6 JavaScript
- **Cloud Deployment & DevOps:** Docker, Docker Compose, Render Cloud Web Service, Git & GitHub